# Wilka Torrico De Carvalho, *Cognitive Scientist and Reinforcement Learning Researcher*


**Contact Information**

*Website:* cogscikid.com
*E-mail:* wcarvalho@g.harvard.edu
*Github:* github.com/wcarvalho
*Google* Scholar


## Employment

**Harvard**, Boston, MA — *Sept 2023 – Present*
Research Fellow

**DeepMind**, London, UK — *Sept 2022 – May 2023*
Research Scientist Intern

**DeepMind**, London, UK — *Mar 2021 – July 2021*
Research Scientist Intern

**Microsoft Research**, Redmond, WA — *June 2018 – Aug 2018*
Research Scientist Intern

**IBM Research**, San Jose, CA — *Sept 2017 – Dec 2017*
Research Scientist Intern

**VISA Research**, Palo Alto, CA — *May 2017 – Aug 2017*
Research Scientist Intern

## Education

**University of Michigan–Ann Arbor**, Ann Arbor, MI — Sep 2018 - July 2023
*School of Engineering*, Ph.D. in Computer Science
Advisors: Satinder Singh, Honglak Lee, Richard Lewis

**University of Southern California**, Los Angeles, CA — Aug 2015 - May 2017
*Viterbi School of Engineering*, M.S. in Computer Science
Advisor: Yan Liu

**Stony Brook University**, Stony Brook, NY — Aug 2011 - May 2015
*College of Arts and Sciences*, B.S. in Physics
Advisor: Axel Drees

**Brooklyn Technical High School**, Brooklyn, NY — May 2007 - May 2011
Diploma in Applied Physics

## Honors & Awards

| | |
|---|---|
| Kempner Institute Neuro-AI Fellowship | 2023 |
| ICML, Neurips Top Reviewer | 2022, 2023 |
| University of Michigan “The Glass is Half-Full” Award | 2019 |
| Hertz Fellowship Finalist (10 % selected) | 2019 |
| 1/200 chosen internationally for Heidelberg Laureate Forum | 2018 |
| GEM National Fellowship sponsored by IBM, Adobe | 2017, 2018 |
| University of Michigan Rackham Merit Fellowship | 2017 |
| ICLR, Neurips BAI Travel Award | 2017, 2019 |
| NSF Graduate Research Fellowship (Neuroscience) | 2015 |
| Provost Award for Academic Excellence (20/5000 graduates chosen) | 2015 |
| Researcher of the Month (1 monthly at Stony Brook University) | 2014 |
| HHMI Minority Undergraduate Research Fellowship | 2014 |
| ΣΠΣ Physics Honor Society (sponsored by Alfred Goldhaber) | 2013 |
| NSF Louis Stokes Alliance for Minority Participation Scholar | 2011 |

Preprints

**Wilka Carvalho**, Sam Hall-McMaster, Honglak Lee, Samuel J. Gershman. *"Preemptive Solving of Future Problems: Multitask Preplay in Humans and Machines."* 2025

**Wilka Carvalho***, Andrew Lampinen*. *"Naturalistic Computational Cognitive Science."* 2025

Aneesh Muppidi, Samuel J. Gershman, **Wilka Carvalho**. *"Unsupervised Agent-centric Representation Learning with Variational Agent Inference."* 2025

Publications

Kunal Jha, **Wilka Carvalho**, Natasha Jaques, Max Kleiman-Weiner. *"Cross-environment Cooperation Enables Zero-shot Multi-agent Coordination."* In ICML, 2025 **(Oral)**

**Wilka Carvalho***, Momchil S Tomov*, William de Cothi*, Caswell Barry, Samuel J. Gershman. *"Predictive representations: building blocks of intelligence."* In Neural Computation, 2024

**Wilka Carvalho**, Andre Saraiva, Angelos Filos, Andrew Lampinen, Loic Matthey, Richard L Lewis, Honglak Lee, Satinder Singh, Danilo Jimenez Rezende, Daniel Zoran. *"Combining behaviors with the successor features keyboard."* In NeurIPS, 2024

**Wilka Carvalho**, Andrew Lampinen, Kyriacos Nikiforou, Felix Hill, Murray Shanahan. *"Feature-Attending Recurrent Modules for Generalization in Reinforcement Learning."* In TMLR, 2023

**Wilka Carvalho**, Angelos Filos, Richard L. Lewis, Honglak Lee, Satinder Singh. *"Composing Task Knowledge with Modular Successor Feature Approximators."* In ICLR, 2023

Lajanugen Logeswaran, **Wilka Carvalho**, Honglak Lee. *"Learning Compositional Tasks from Language Instructions."* In AAAI, 2023

Chris Hoang, Sungryull Sohn, Jongwook Choi, **Wilka Carvalho**, Honglak Lee. *"Successor Feature Landmarks for Long-Horizon Goal-Conditioned Reinforcement Learning."* In NeurIPS, 2021

**Wilka Carvalho**, Anthony Liang, Kimin Lee, Sungryull Sohn, Honglak Lee, Richard L. Lewis, Satinder Singh. *"Reinforcement Learning for Sparse-Reward Object-Interaction Tasks in First-person Simulated 3D Environments."* In IJCAI, 2021

**Wilka Carvalho***, Sanjay Purushotham*, Tanachat Nilanon, Yan Liu. *"Variational Recurrent Adversarial Domain Adaptation."* In ICLR, 2017

Workshop Publications

**Wilka Carvalho**, Vikram Goddla, Ishaan Sinha, Hoon Shin, Kunal Jha. *"NiceWebRL: a Python library for human subject experiments with reinforcement learning environments."* In Human-AI Complementarity for Decision Making Workshop, 2025 **(Oral)**

Lajanugen Logeswaran, **Wilka Carvalho**, Honglak Lee. *"Learning Compositional Tasks from Language Instructions."* In NeurIPS Deep RL Workshop, 2021

**Wilka Carvalho**, Murray Shanahan. *"Learning to Represent State with Perceptual Schemata."* In ICML {Unsupervised RL, Real Life RL} Workshops, 2021

**Wilka Carvalho**, Anthony Liang, Kimin Lee, Sungryull Sohn, Honglak Lee, Richard L. Lewis, Satinder Singh. *"Reinforcement Learning for Sparse-Reward Object-Interaction Tasks in First-person Simulated 3D Environments."* In {ICML Object-Oriented Learning, NeurIPS DeepRL} Workshops, 2020

Chris Hoang, Sungryull Sohn, Jongwook Choi, **Wilka Carvalho**, Honglak Lee. *"Successor Land-*

*marks for Efficient Exploration and Long-Horizon Navigation."* In NeurIPS DeepRL Workshop, 2020

**Wilka Carvalho**, Kimin Lee, Anthony Liang, Ryan Krueger, Richard L. Lewis, Satinder Singh, Honglak Lee. *"Efficiently Learning to Perform Household Task with Object-oriented Exploration."* In NeurIPS BAI, 2019 **(Oral)**

Bryant Chen*, **Wilka Carvalho***, Benjamin Edwards, Taesung Lee, Ian Molloy, Heiko Ludwig. *"Detecting Backdoor Attacks on Deep Neural Networks by Activation Clustering."* In AAAI Artificial Intelligence Safety Workshop, 2018 **(Best Paper)**

Sanjay Purushotham*, **Wilka Carvalho***, Yan Liu. *"Variational Adversarial Deep Domain Adaptation for Health Care Time Series Analysis."* In NeurIPS Machine Learning for Healthcare Workshop, 2016 **(Spotlight)**

**Wilka Carvalho**. *"Modeling a Detection of internally reflected Cherenkov light (DIRC) Particle Detector for High-Multiplicity Collisions."* In SURC, 2015

## Talks

**NiceWebRL: A Python Library for Human Experiments with Machine RL Environments**
Human-AI Complementarity for Decision Making Workshop. *Carnegie Mellon University.* (September, 2025)
Cognitive Computational Neuroscience Conference. *Amsterdam.* (August, 2025)

**Naturalistic Computational Cognitive Science**
Cognitive Computational Neuroscience Conference. *Amsterdam.* (August, 2025)

**Scaling Up Reinforcement Learning Theories of Cognition**
Cognition, Brain, and Behavior Seminar. *Harvard University.* (April, 2025)
Center for Theoretical Neuroscience. *Columbia University.* (April, 2025)
Mattar Lab. *NYU.* (April, 2025)
Emerge Lab. *NYU.* (April, 2025)
Computation and Decision-Making Lab. *NYU.* (April, 2025)
RL Seminar. *Tübingen.* (March, 2025)
Google DeepMind. *San Francisco.* (March, 2025)
Cognition Seminar. *UC Berkeley.* (March, 2025)
The Rational Altruism Lab. *UCLA.* (March, 2025)
Social and Decision Neuroscience Seminar. *Caltech.* (March, 2025)
Symposium on "Frontiers of Machine Learning and AI. *USC.* (March, 2025)
Neurolunch Seminar. *Harvard.* (March, 2025)
RL Seminar. *UMich.* (February, 2025)
Vision Lab. *Harvard University.* (February, 2025)
Sabatini Lab. *Harvard University.* (February, 2025)

**Predictive Representations: Building Blocks of Intelligence**
Computational Cognitive Science Laboratory. *UC Berkeley.* (March, 2025)
Computational Cognitive Neuroscience Lab. *Harvard University.* (February, 2025)
Neuro Systems Club. *Harvard University.* (Fall, 2024)
Mathis Group for Computational Neuroscience and AI. *EPFL.* (Fall, 2024)
Theory of Learning Lab. *Gatsby Computational Neuroscience Unit (UCL).* (Summer, 2024)
Conference on the Mathematics of Neuroscience and AI. *Rome, Italy.* (Summer, 2024)
Theoretical Cognitive Neuroscience Lab. *Boston University.* (Fall, 2023)

**Towards Human-like Deep Reinforcement Learning**

Computational Cognitive Science Group. *Princeton.* (Fall, 2023)
Kempner Institute. *Harvard University.* (March, 2023)

**Composing Task Knowledge with Modular Successor Feature Approximators**
RL at Harvard. *Harvard University.* (August, 2023)

**Weak Inductive Biases for Composable Primitive Representations**
Computational Neuroscience Group. *UCL.* (October, 2022)
Intelligent Robot Lab. *Brown.* (February, 2022)
Computational Cognitive Neuroscience Lab. *Harvard University.* (February, 2022)
Computational Cognitive Neuroscience Lab . *University of California—Berkeley.* (January, 2022)
Deep Learning Community of Practice. *Intel.* (January, 2022)
Deep RL Workshop. *NeurIPS.* (December, 2021)

**Reinforcement Learning for Sparse-Reward Object-Interaction Tasks in a First-Person Simulated 3D Environment**
Object Representations for Learning Workshop. *NeurIPS.* (December, 2020)
Cognitive Tools Lab. *University of California—San Diego.* (November, 2020)
Cognitive Science Community. *University of Michigan.* (November, 2020)
Stony Brook Society of Physics. *Stony Brook, NY.* (April, 2020)

**Variational Recurrent Adversarial Deep Domain Adaptation**
Machine Learning Lunch Seminar. *University of Southern California.* (April, 2017)

## Patents

**Wilka Carvalho**, Angelos Filos, Danilo Jimenez Rezende, Daniel Zoran. *"Controlling Agents by Transferring Successor Features To New tasks."* 2023

**Wilka Carvalho**. *"Compositional Generalization in Reinforcement Learning."* 2023

Bryant Chen, **Wilka Carvalho**, Heiko Ludig, Ian Molloy, Jialong Zhang, Benjamin Edwards. *"Detecting poisoning attacks on neural networks by activation clustering."* 2020

**Wilka Carvalho**, Bryant Chen, Benjamin Edwards, Taesung Lee, Ian Molloy, Jialong Zhang. *"Using Gradients to Detect Backdoors in Neural Networks."* 2018

**Wilka Carvalho**, Yan Liu, Tanachat Nilanon, Sanjay Purushotham. *"Effective Knowledge Transfer Among Patient Populations via Deep Learning."* 2017

## Research Experience

**University of Michigan–Ann Arbor**, Ann Arbor, Michigan USA
**AI Lab**, *August 2018 – Present*
Advisors: Honglak Lee, Satinder Singh, Richard Lewis

- Studying how deep reinforcement learning agents can discover representations and temporally-extended behaviors that enable generalization

**DeepMind**, London, UK
**Generative Models Team**, *September 2022 – May 2023*
Advisor: Danilo Rezende

- Studying transfer in 3D environments with successor features

**DeepMind**, London, UK
**Cognition Team**, *March 2021 – July 2021*
Advisor: Murray Shanahan

- Studying agent-state representations for generalization of deep reinforcement learning agents in 3D environments

**Microsoft Research**, Redmond, Washington USA
**Medical Devices Group**, *June 2018 – August 2018*
Advisor: Sumit Basu

- Developed a machine learning model that can predict clinical measures from physiological signals measured by a wearable device.

**IBM Research**, San Jose, California USA
**AI Platform Research Group**, *September 2017 – December 2017*
Advisor: Heiko Ludwig

- Contributed to novel research algorithm by suggesting subspace projection technique that increased our performance from 15% to 95% accuracy.
- Research led to workshop publication and was presented in global meeting to IBM executives including CEO and selected for funding.

**Visa Research**, Palo Alto, California USA
**Data Analytics Group**, *June 2017 – August 2017*
Advisor: Hao Yang

- Implemented model and baselines for language generation and question answering.

**University of Southern California**, Los Angeles, California USA
**Melady Machine Learning Lab**, *November 2015 – May 2017*
Advisor: Yan Liu
Samsung and NSF funded project: "Variational Adversarial Deep Domain Adaptation for Health Care Time Series Analysis"

- Implemented novel neural network that employed variational inference and adversarial training for transfer learning of multivariate time-series.
- Research led to a publication and a patent.
- Communicated research to general public through research feature by the USC Graduate School and to technical audience at ICLR poster presentation.

**Stony Brook Univeristy**, Stony Brook, New York USA
**Heavy Ion Research Group**, *January 2013 – August 2015*
Advisor: Axel Drees
DOE funded project: "Modeling a Detection of internally reflected Cherenkov light Particle Detector for High-Multiplicity Collisions"

- Contributed methods from multivariate calculus and linear algebra to particle detection algorithm. Accuracy improved from 60% to 80%.
- Designed and implemented a statistical analysis pipeline in C++ for measuring efficacy of particle detection algorithm.

**Stony Brook Univeristy**, Stony Brook, New York USA
**Computational Neuroscience Group**, *Fall 2014*
Advisor: Giancarlo La Camera
NSF LSAMP funded project: "Spectral Analysis of Rodent Neural Data"

- Performed spectral analyses on neural data to determine behavioral correlates of neural activity.

**California Institute of Technology**, Pasadena, California USA
**Emotion and Social Cognition Laboratory**, *Summer 2014*

Advisor: Ralph Adolphs
HHMI funded project: "Modeling the Cognitive Process of Attributing Traits to Others"

- Formulated a trait learning behavioral experiment to study human inference.

**University of Minnesota**, Minneapolis, Minnesota USA
**Neuromodulation Research and Technology Laboratory**, *Summer 2013*
Advisor: Matthew Johnson
NIH funded project: "Modeling Deep Brain Stimulation of Globus Palidus Internus"

- Implemented python script to build a biologically feasible computational model of neural networks

**National Central University**, Jhongli City, Taiwan
**Turbulent Combustion Laboratory**, *Summer 2012*
Advisor: Shenqyang Shy

## Teaching Experience

**Stony Brook University**, Stony Brook, NY
**Calculus Instructor**, *Spring 2015*
Worked with two math professors to develop and teach a supplementary calculus curriculum that promoted minority representation in stem majors.

**Stony Brook University**, Stony Brook, NY
**Educational Opportunity Program Personal Tutor**, *Spring 2013 - Fall 2014*
Tutored marginalized students in introductory physics and math courses

## Service

Action Editor, Transactions on Machine Learning Research, 2023
Area Chair, Reinforcement Lerning Conference, 2023
Reviewer, Neural Computation, 2023
Reviewer, NeurIPS, 2020, 2021, 2022, 2023
Reviewer, ICLR, 2021, 2022
Reviewer, ICML, 2021, 2022
Student Volunteer, ICLR, 2017

## Outreach

*HS2 Engineer's Highschool Panel*, Virtual, 2021
*Stony Brook Society of Physics*, Stony Brook, NY, 2020
*Michigan Explore Graduate Studies Workshop*, Ann Arbor, MI, 2019
*Research and Fellowships Week NSF Panel*, Los Angeles, CA, 2016
*National Society of Black Engineers Grad Panel*, Los Angeles, CA, 2016
*Graduate School External Fellowship Boot Camp*, Los Angeles, CA, 2016
*Mentored marginalized high school youth through the Pullias Center for Higher Education*, Los Angeles, CA, 2016
*Engineering Graduate Diversity Symposium*, Los Angeles, CA, 2015
*Black Student Association: What it takes to go to Graduate School*, Los Angeles, CA, 2015
*Collegiate Science and Technology Entry Program Undergraduate Research Panel*, Stony Brook, CA, 2014

## Skills

Machine Learning Software: Pytorch, TensorFlow, Theano, Keras
Neuroscience Software: Neuron
Languages: Python, C++, C, Java
Systems: Unix, Linux, OSX

## Press

Finding Gaps in Your Grad School Apps
This Week In Machine Learning (TWIML) Research Feature
University of Michigan Research Feature
Exploring the source of social stereotypes
Black History Month: Why a career in science?
Research Feature by the USC Graduate School
2015 NSF Graduate Research Fellow Wilka Carvalho
Biomath Learning Center Launches Modified Supplemental Instruction Program
URECA Research of the Month: Wilka Carvalho
Student Feature by Stony Brook University

## Interests

• traveling • chess • software development • improvisational dance • deadpan humor